INCH-POUND

MIL-DTL-16377/79
18 December 2008

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, GENERAL LIGHTING, 34 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 77.4T8, 77.5T8, 344.2T8, AND 344.3T8

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

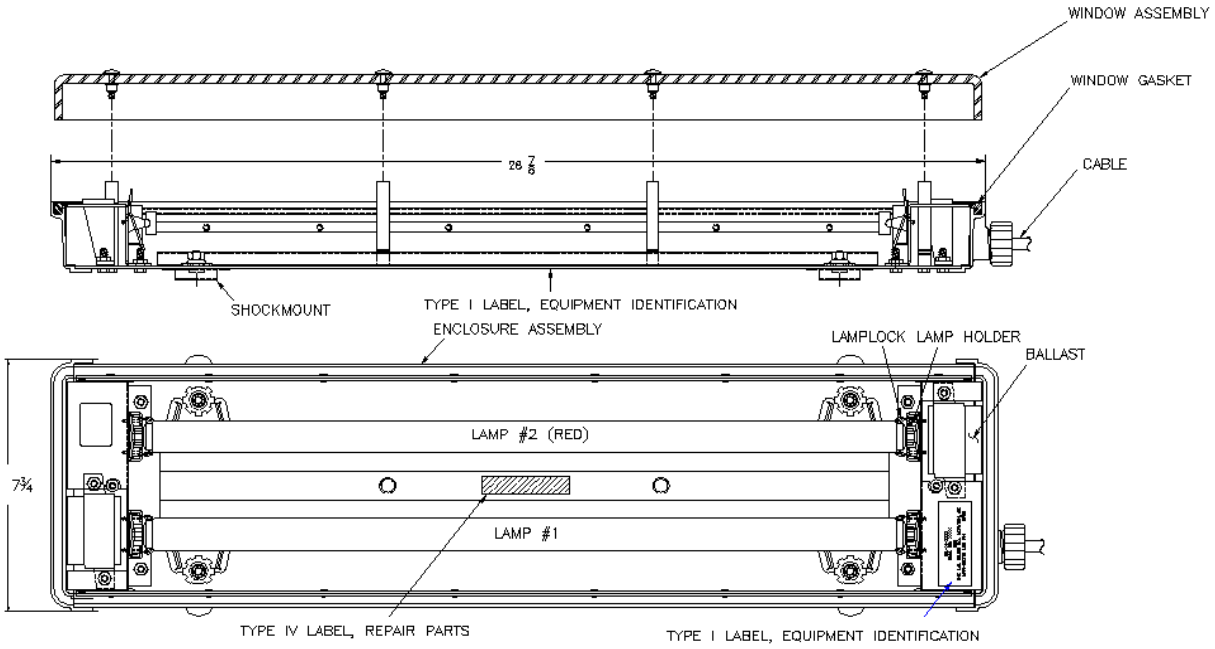


FIGURE 1. Typical view of symbols 77.4T8 and 344.2T8, symbols 77.5T8, and 344.3T8 use identical construction with flush mounting flange not shown.

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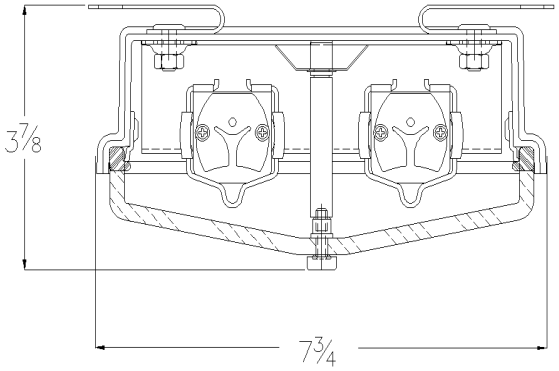
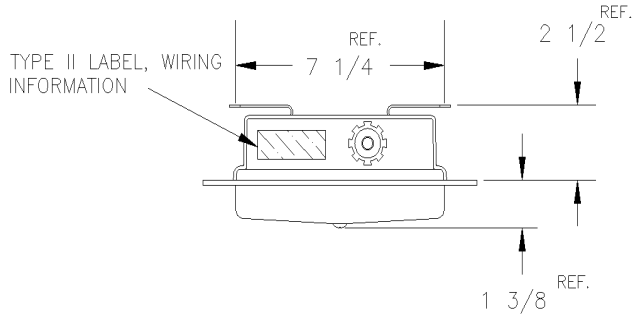


FIGURE 2. Typical section for symbols 77.4T8 and 344.2T8.



NOTE:

1. Type II label required for all fixtures.

FIGURE 3. Typical section for symbols 77.5T8 and 344.3T8.

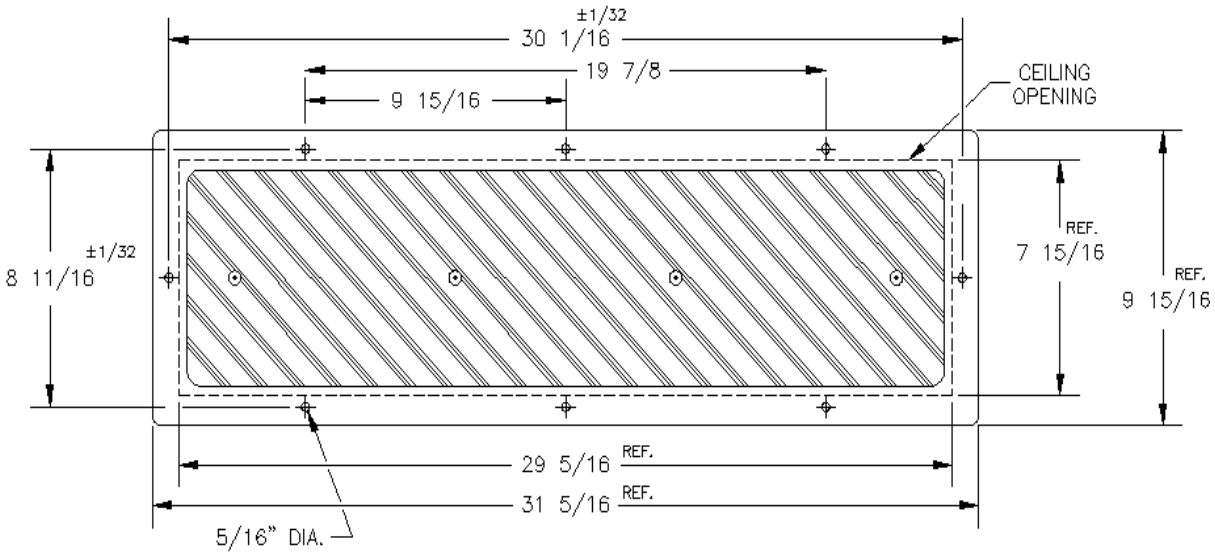


FIGURE 4. Flush mounting dimension for symbols 77.5T8 and 344.3T8.

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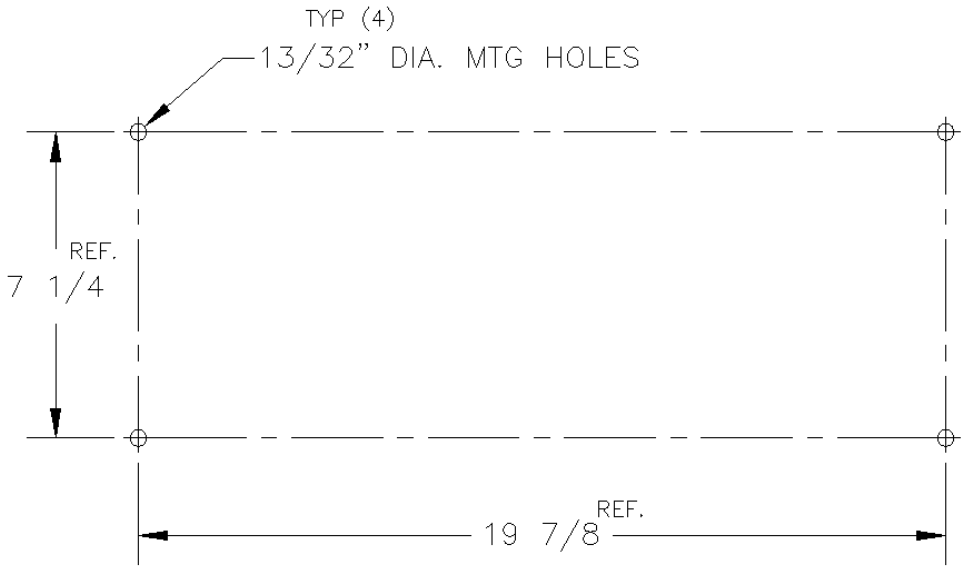


FIGURE 5. Standard mounting dimensions utilizing shock mounts for symbols 77.4T8 and 344.2T8.

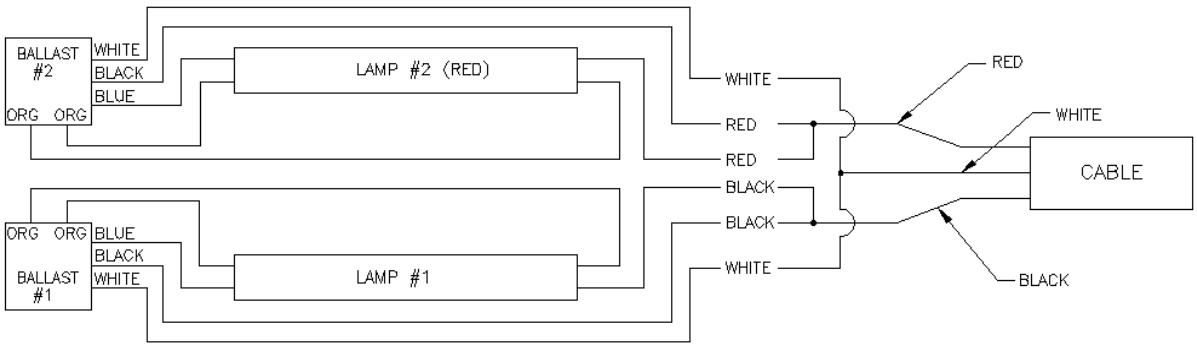


FIGURE 6. Typical wiring diagram.

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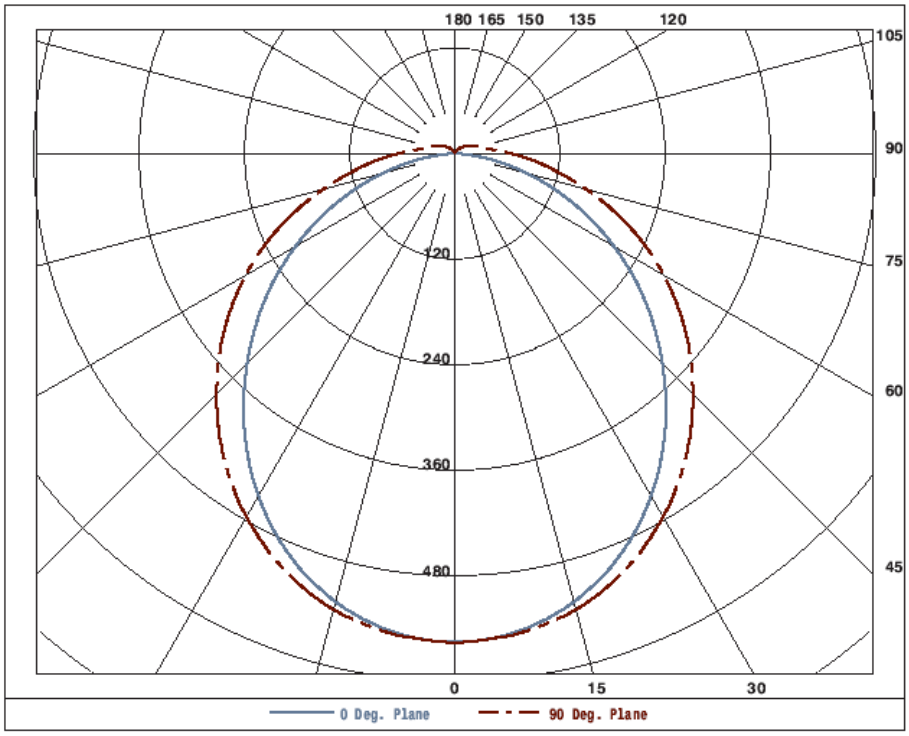


FIGURE 7. Candela distribution curves, white diffusing window, symbols 77.4T8 and 77.5T8.

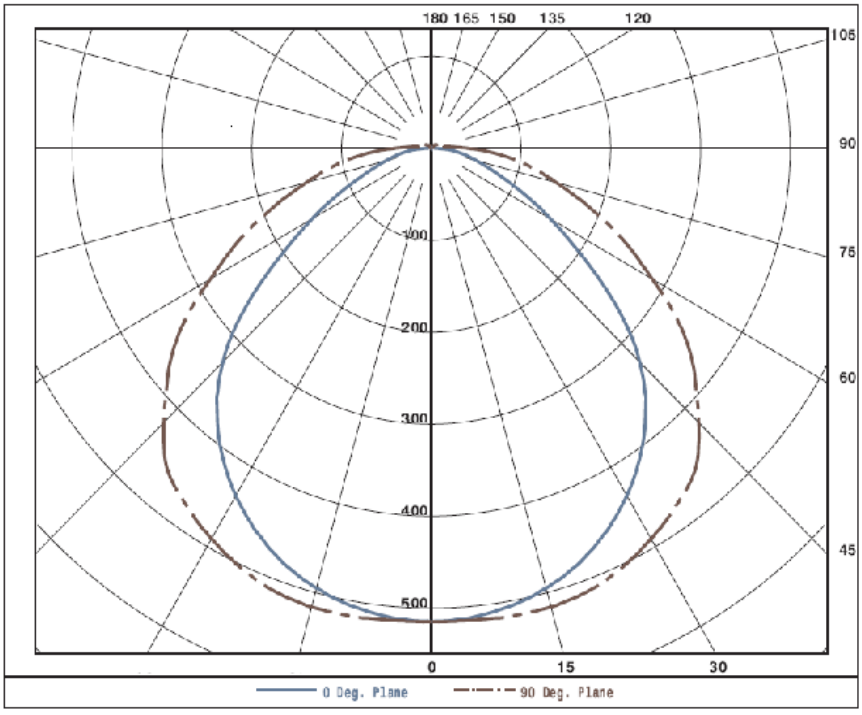


FIGURE 8. Candela distribution curves, prismatic window, symbols 344.2T8 and 344.3T8.

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TABLE I. Numerical candela distribution, white diffusing window, symbols 77.4T8 and 77.5T8.

Vertical angle	Candela distribution	
	0-degree	90-degree
0	556	556
5	552	552
15	528	537
25	480	501
35	413	449
45	335	385
55	253	314
65	167	236
75	85	155
85	18	90
90	4	65
95	4	46
105	3	30
115	2	21
125	2	14
135	1	8
145	0	4
155	0	0
165	0	0
175	0	0
180	0	0

TABLE II. Numerical candela distribution, prismatic window, symbols 344.2T8 and 344.3T8.

Vertical angle	Candela distribution	
	0-degree	90-degree
0	515	515
5	512	515
15	495	516
25	461	503
35	407	475
45	328	421
55	207	340
65	109	236
75	52	144
85	19	75
90	5	40
95	4	20
105	3	13
115	2	11
125	1	6
135	1	3
145	0	2
155	0	0
165	0	0
175	0	0
180	0	0

PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other IESNA procedures.

Plane: 0 degrees is parallel to lamps, 90 degrees is perpendicular to lamps.

Electrical values: 120 volts, 60 hertz, 36.1 watts input power.

Ambient temperature: 25±1 °C.

Lamps: GE F17T8-SP41-ECO (seasoned for at least 100 hours).

Lamps correction: lamp rated at 1325 lumens.

Ballast: furnished with fixture, see MIL-DTL-16377/81 for specification.

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REQUIREMENTS:

CLASSIFICATION:

Type: I (Fluorescent).

Class: 2 (General).

SYMBOL NUMBERS:

77.4T8: Complete lighting fixture assembly, standard mounting, with two 17-watt T8 cool white fluorescent lamp and white diffusing window assembly.

77.5T8: Complete lighting fixture assembly, flush mounting, with two 17-watt T8 cool white fluorescent lamp and white diffusing window assembly.

344.2T8: Complete lighting fixture assembly, standard mounting, with two 17-watt T8 cool white fluorescent lamp and clear and white prismatic window assembly.

344.3T8: Complete lighting fixture assembly, flush mounting, with two 17-watt T8 cool white fluorescent lamp and clear and white prismatic window assembly.

DESIGN:

Electrical rating:	34 watts, 120 volts, 60 hertz, ungrounded electrical distribution systems. Total input power: 36.1 watts at 120 volts.
Dimension and configurations:	See figures 1, 2, and 3.
Photometric data:	See figures 7 and 8 and tables I and II.
Wiring diagram:	See figure 6.
Ballast:	See MIL-DTL-16377/81 for specification.
Mounting dimensions:	
Symbols 77.4T8 and 344.2T8:	Standard mounting (surface), see figure 5.
Symbols 77.5T8 and 344.3T8:	Flush mounting (recessed), see figure 4.
Window assembly:	
Symbols 77.4T8 and 77.5T8:	White window, see table III, and figure 27 of MIL-DTL-16377.
Symbols 344.2T8 and 344.3T8:	Prismatic window, see table III, and figure 27 of MIL-DTL-16377.
Enclosure:	Watertight; see 3.5.11 of MIL-DTL-16377.
Cable:	10 feet (minimum) of type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 77.4T8 and 344.2T8	Not to exceed 12 pounds.
Symbols 77.5T8 and 344.3T8	Not to exceed 13 pounds, 13 ounces.

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DESIGNATION AND MARKING:

The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I labels, equipment identification, reference MIL-DTL-16377 figure 11, shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.

b. One Type II label, circuit wiring information, reference MIL-DTL-16377 figure 12, located on outside surface adjacent to cable entrance.

c. One Type IV label, repair parts, reference MIL-DTL-16377 figure 14, information to be shown shall include “USE THE FOLLOWING REPAIR PARTS ONLY.” List all standard parts shown in table III, located on an inner nonreflecting surface.

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:

Complete fixtures:	See table IV.
Window assemblies:	See table V.
Comparison inspection:	See tables IV and V.
Conformance inspection:	See MIL-DTL-16377.

Inspection of product for delivery:

Examination	See MIL-DTL-16377.
Conformance test	See MIL-DTL-16377.
Examination for preparation of delivery	See MIL-DTL-16377.

GENERAL INFORMATION:

Table III. Fixture standard parts.

Table IV. Complete fixture first article tests.

Table V. Window assembly first article tests.

Table VI. Complete fixture part numbers.

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TABLE III. Standard parts.

Name	Military p/n or specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F17T8/CW	6240-01-391-6027	---
Lamplock	M16377/40-003	---	---
Lampholder	UL 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001	---	Integrated starter type
Shockmount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, diffusing	M16377/11-001	6210-00-399-7035	1/
Window assembly, prismatic	M16377/11-003	6210-01-169-5997	1/
Filter, 17-watt T8 fluorescent lamp (DARK ADAPTATION)	M16377/42-010	---	Red transparent

NOTE:
1/ Window assemblies (diffusing and prismatic) are the same as T12 lamp fixtures (MIL-DTL-16377/11).

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TABLE IV. First article tests and comparison inspection for complete fixtures.[1/]

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Electromagnetic interference	3.6.14	4.8.16	---
Electrical equipment interface testing requirements: MIL-STD-1399-300	MIL-STD-1399-300	MIL-STD-1399-300	[2/], [3/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/]
Shock	3.5.12	4.8.7	[4/]
Vibration	3.5.13	4.8.8	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Noise	3.5.14	4.8.9	---
Continuity of grounding	3.6.13.4	4.8.12	---
Light output	3.7.1	4.8.6.1	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTES:
1/ Requirements and test methods use MIL-DTL-16377.
2/ If less, then 1 KVA exception not applicable for input waveform THD.
3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300.
4/ Tests shall be performed in the order specified.

TABLE V. First article tests and comparison inspection for window assemblies.[1/]

Test	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/]
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
1/ Testing requirements and methods use MIL-DTL-16377.
2/ Tests shall be performed in the order specified.

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TABLE VI. Complete fixture part numbers.

Name	Military p/n
Complete fixture, symbol 77.4T8	M16377/79-77.4T8
Complete fixture, symbol 77.5T8	M16377/79-77.5T8
Complete fixture, symbol 344.2T8	M16377/79-344.2T8
Complete fixture, Symbol 344.3T8	M16377/79-344.3T8

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Review Activities:
Navy – AS, CG
Air Force – 03
DLA – GS2
GSA – FAS

Preparing Activity:
Navy – SH
(Project 6210-2006-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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